

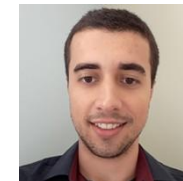
# Capacity Expansion Planning Under the Risk of Hurricanes: An Analysis of the NC Electric Power System

Victor Faria, Matteo Nicoli, Anderson R. Queiroz

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[vaduraes@ncsu.edu](mailto:vaduraes@ncsu.edu)  
[matteo.nicoli@polito.it](mailto:matteo.nicoli@polito.it)  
[ardequei@ncsu.edu](mailto:ardequei@ncsu.edu)



# The Capacity Expansion Planning Problem

- Find the optimal combination of system resources (size and deployment period) to minimize the cost of meeting future energy demands.

## Represent Existing Power System

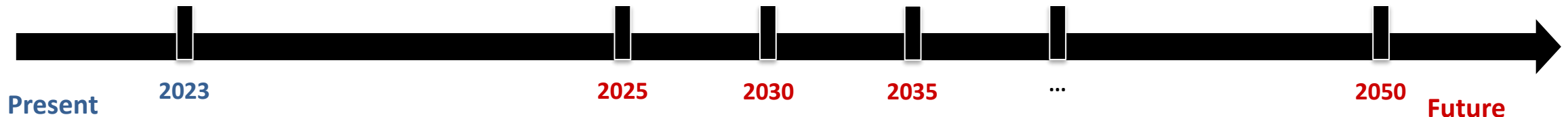
- Technologies in the System (NG, COAL, SOLAR, WIND, BATTERY, HYDRO, ...)
- What is the Installed Capacity of Each Technology Type
- Remaining Technology Life
- Technology Efficiency
- Technology Emission Rate
- Technology Location, ...

## Represent Future Potential Technologies And Their Characteristics

- Improvements in Cost/Efficiency due to Tech Maturity
- Fuel Price Projections
- Retrofit Options For Existing NG/Coal Plants
- CO2 Capture Technologies, ...

## System Constraints

- Energy Reserve Requirements (Maintain Safe Operation in Critical Conditions)
- Maximum Resource Available (Biomass, Wind, ...)
- Emission Targets, Flow Constraints, ...



# Looking at Hurricane Damage to Power Systems

- The estimated cost to North Carolina customers for the storm repairs made during 2018-2019 due to Hurricanes Florence, and Dorian account for more than **US\$1 billion** (Duke Energy 2021)
- Hurricane Irma (2017) is estimated to cost **US\$1.3 billion** to Florida Power & Light customers (FPL 2018). Hurricane Ian (2022) **US\$1.1 billion** (NextEra 2022)

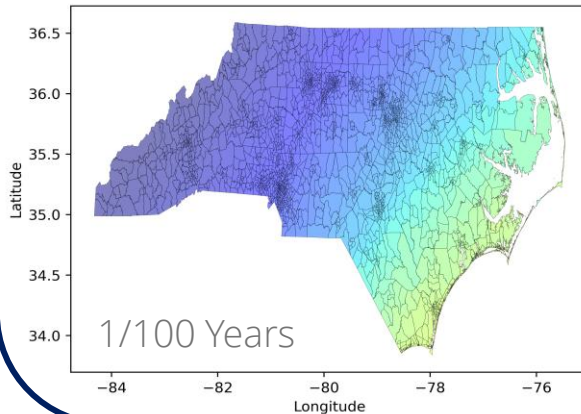
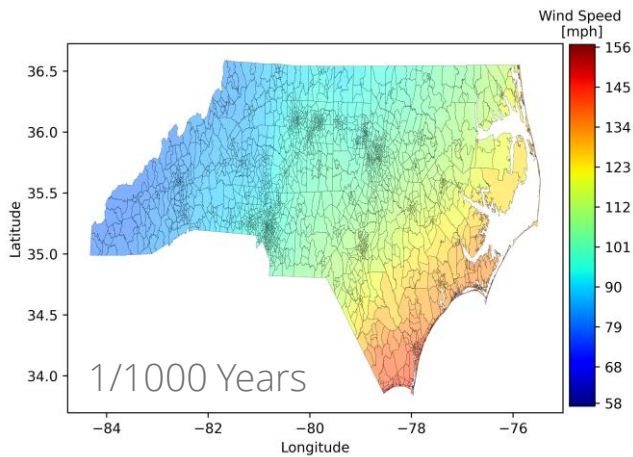
WHY NOT MODEL THESE DAMAGES ON OUR MODELS?



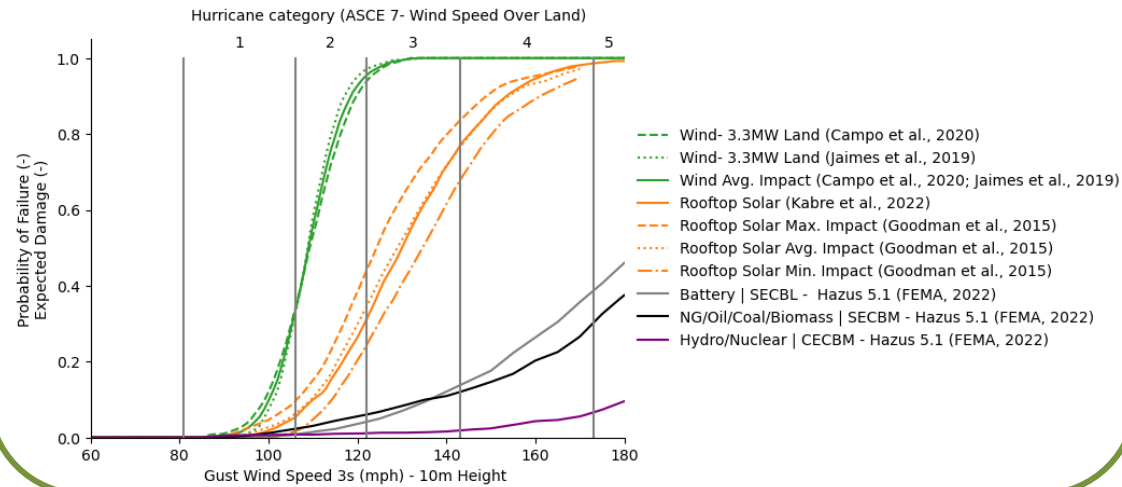
# How to Model Hurricane Damage?

Extreme Wind Speeds From FEMA HAZUS Model (peak gust at 10m).

Different Recurrences:  
20, 50, 100, 200, 500, 1000 yrs.



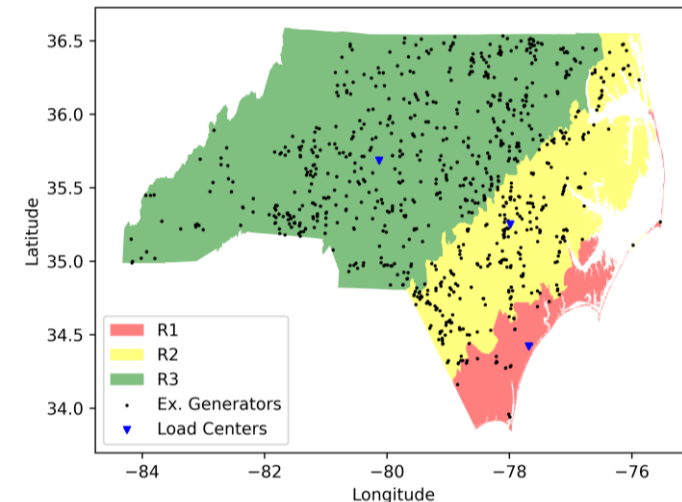
## Fragility Curves



**DAMAGE=Frangility (Wind Statistics)**

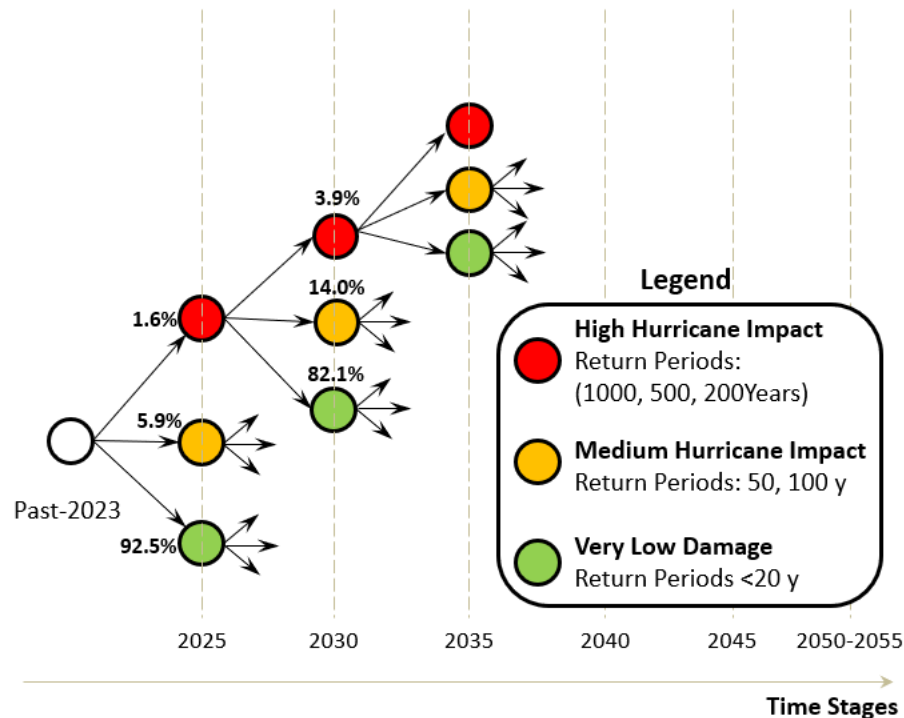
**DAMAGE** for each technology type for each county in the region of interest and for different hurricane conditions

## Regionalization Adopted



# Stochastic Optimization

Definition of the Problem Scenario Tree (All Mapped Outcomes):



- Scenarios:
- High Hurricane Impact (At Least One 1000y, 500y or 200y Event Happened)
  - Medium Hurricane Impact (At Least One 100y or 50y Event Happened)
  - Low Hurricane Impact (No Hurricane or Hurricanes Bellow 20y Happened)

In each Scenario, Damage to Infrastructure is Compute as The Expected Damage, Conditioned on the Hurricane Intensities (Remember This is Technology and Region Dependent)

After the realization of a hurricane scenario, the existing capacity of any technology is reduced proportionally to its expected damage.

Exponential Growth of Scenarios: Total # Scenarios =  $3^6 = 729$



# Stochastic Optimization

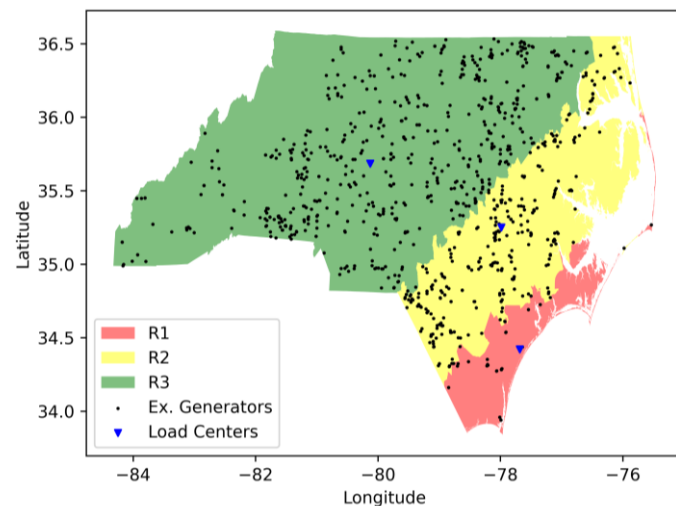
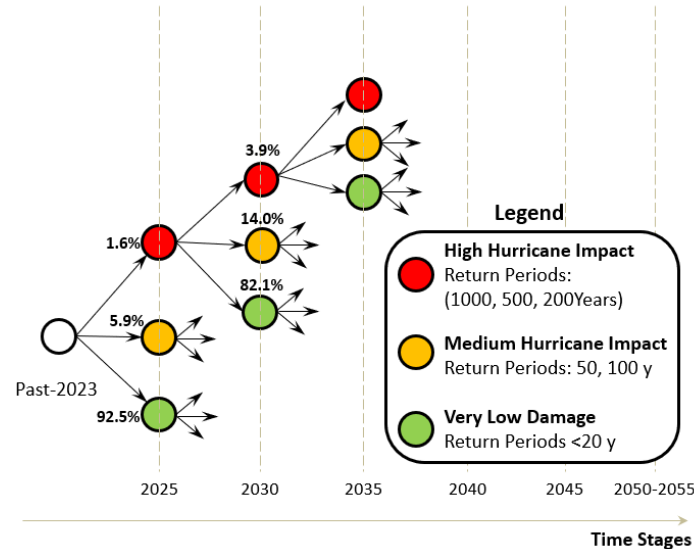
Given different extreme wind scenarios with their corresponding probabilities, a stochastic optimization looks at each time step for the set of solutions (deployment size and type) that minimizes the **expected cost** to meet energy demand throughout the entire time horizon.

## Deterministic vs. Stochastic

**Deterministic:** Assumes the Future Is Known (One Possible Realization- e.g., no Hurricanes)

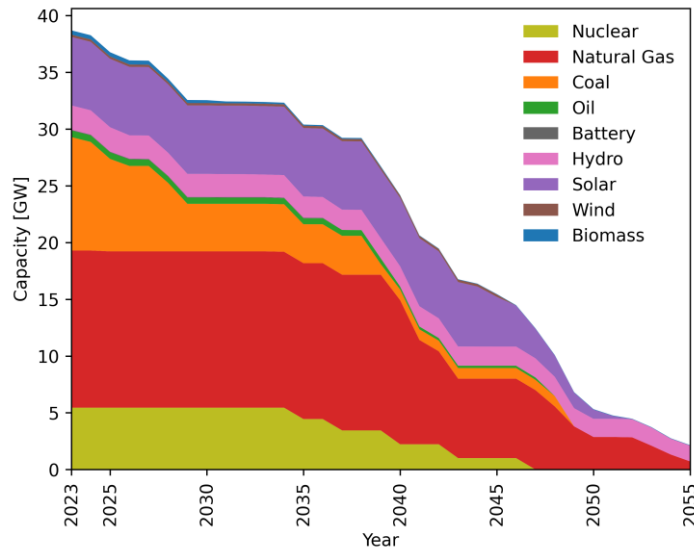
**Stochastic:** Represents The Future as a Set of Possible Realizations (Scenarios) With Their Respective Probabilities.

By Minimizing the Expected Costs Across Different Scenarios, Stochastic Models can Provide Strategies to Hedge Against Undesirable Economic Losses.



# The North Carolina Case

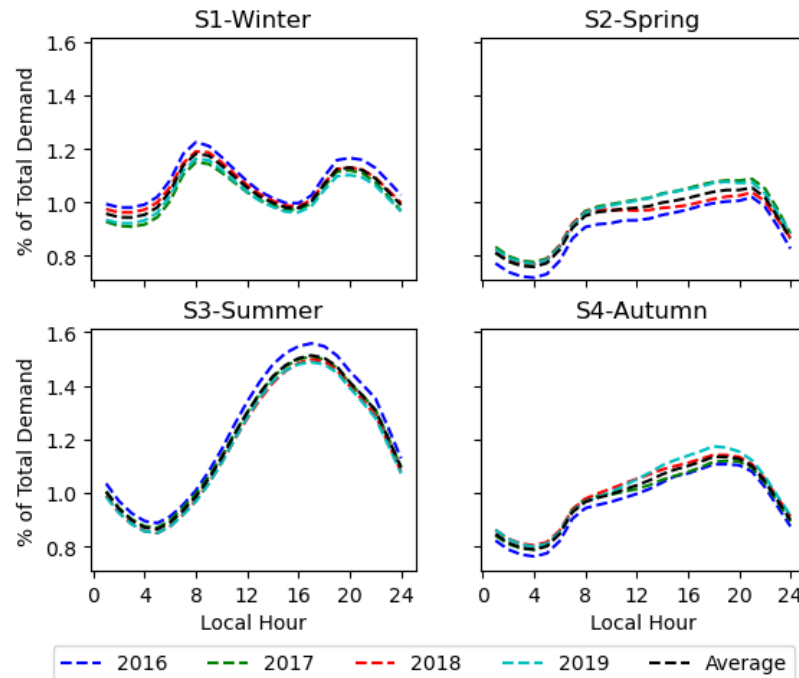
Fig 1. Total Capacity for Existing Technologies. Capacity Decreases as Existing Technologies Retire.



- About 36% Capacity From NG
- 25% From Coal (Many Retirements Soon)
- 15% Solar
- 14% Nuclear
- 5% Hydro
- ~5% (Biomass, Land-Wind, ...)

>85% of Total Electricity Generated by NG, Coal and Nuclear

Fig 2. Load Distribution on NC



On NC the critical load usually happens during the early winter mornings. Leading to the enforcement of a 17% energy reserve requirement.

- Solar energy has a very low contribution to meeting this reserve (Low Capacity Credit)

Fig 3. CF Distribution For Utility and Distributed PV

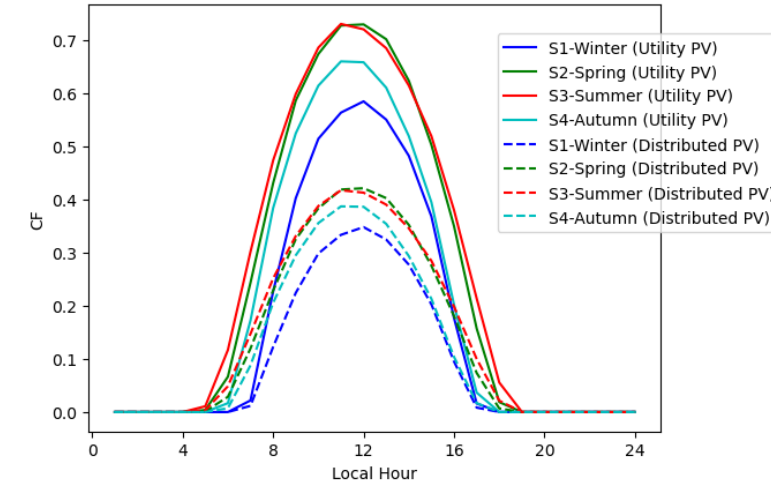
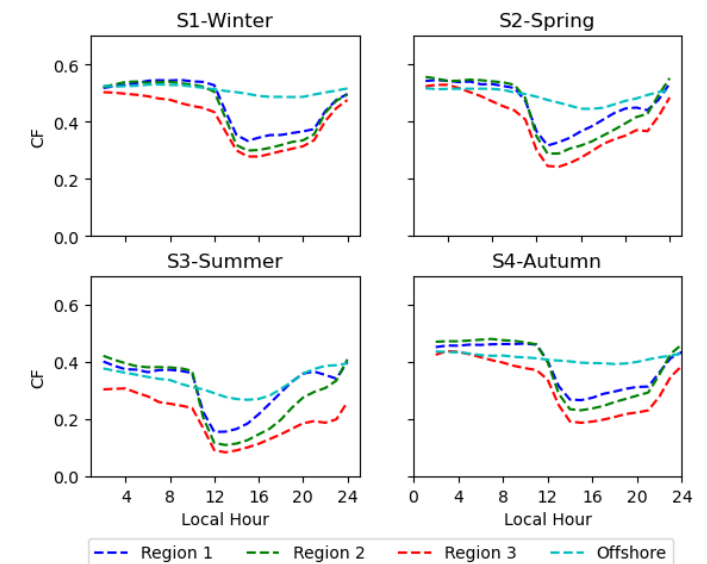


Fig 4. CF Distribution For Wind Technologies



# The North Carolina Case

Fig 5. Evolution of CF for Utility and Distributed PV

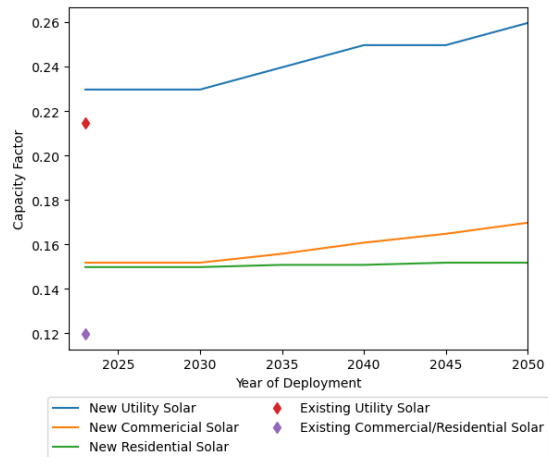
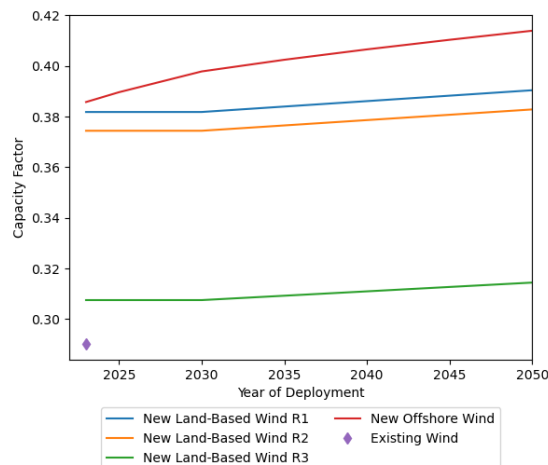


Fig 6. Evolution of CF for Wind Technologies



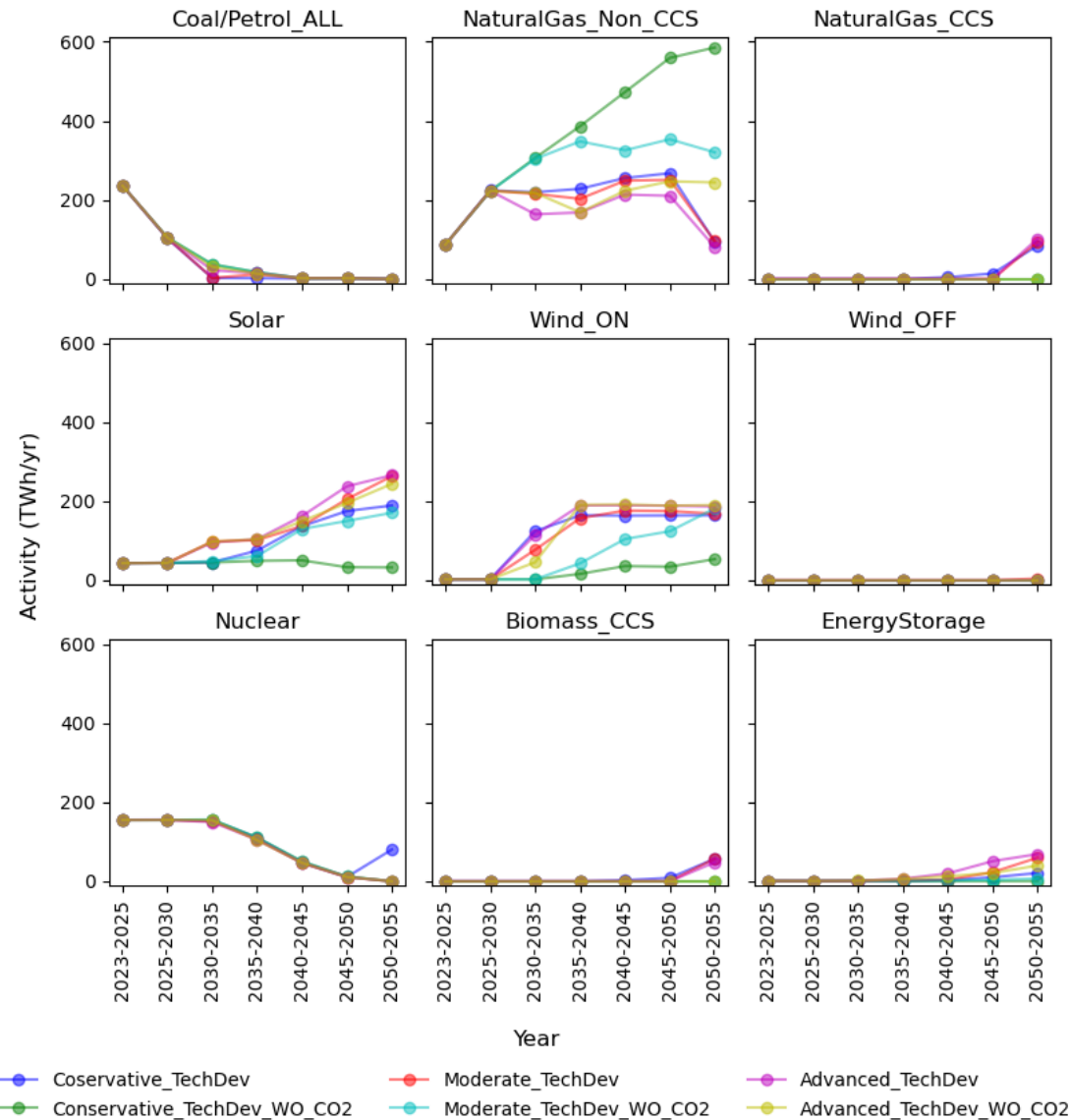
## Important Limits:

- **Emission Limits** (House Bill 951, 2021):
  - 70% CO2 reduction from 2005 levels by 2030
  - Carbon neutrality by 2050
- **Max Capacity For Wind** In Each Region (NREL)
- **Max Available Biomass** (Billion-Ton Report, DOE)

## Other Information:

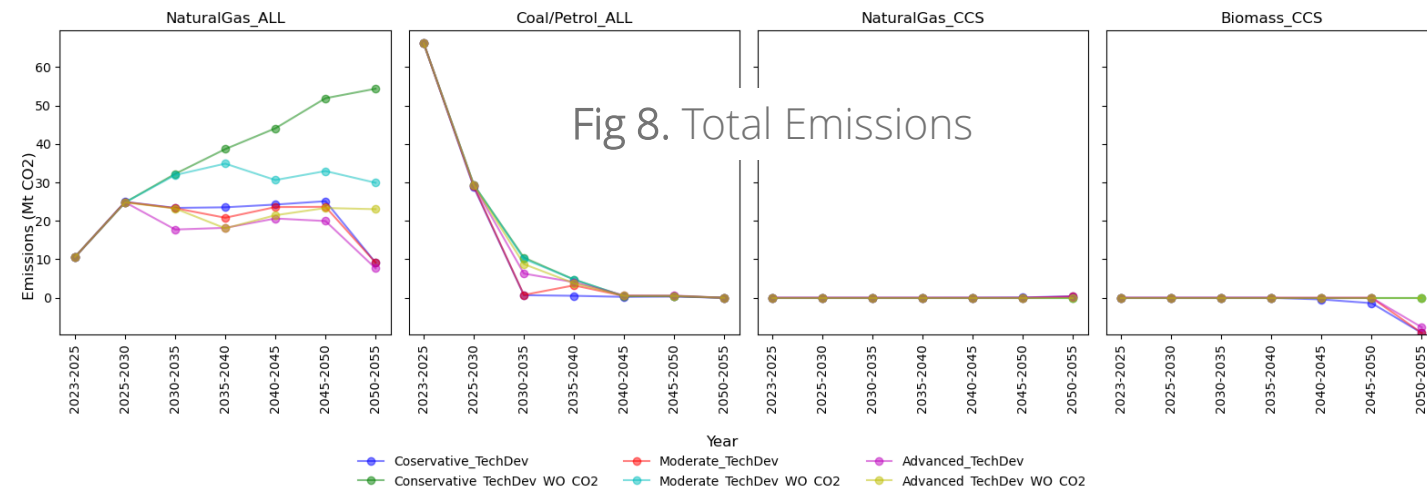
- Most of our **costs** were extracted from the Annual Technology Baseline (**ATB**), and EIA reports (**AEO**).
- **Emissions** from a combination of **EIA** (Existing Tech) and **ATB** reports (New Tech).
- Many reports from **Duke Energy** (Largest Utility in NC) were used to project **load growth**, **distributed PV growth**, and to inform **planned generation retirements** (Expected Remaining Tech Life).

Fig 7. Total Activity for Different Technologies Under Conservative, Moderate, and Advanced Cost Assumptions from NERL-ATB. With and W/O CO2 Targets



### Some Observations:

- Four Seasons and One Representative Day Per Season (24h)
- Simulations With **Conservative/Moderate/Advanced Tech Development** from NREL-ATB; Simulations With and Without CO2 Targets
- In all configurations, **coal loses its participation in the total generation**
- **With CO2 targets**, Conservative/Mod/Advanced cases do **not change significantly** from each other (Biggest Changes are in Storage)
- **Without CO2 targets**, the participation of **NG increases** very fast in the **conservative** case, taking space from renewables.
- The **Advanced** case **W/O CO2 targets** behave similarly with exception of 2050-2055
- **Biomass** working as an **important tech** in **meeting 2050 CO2 Target**

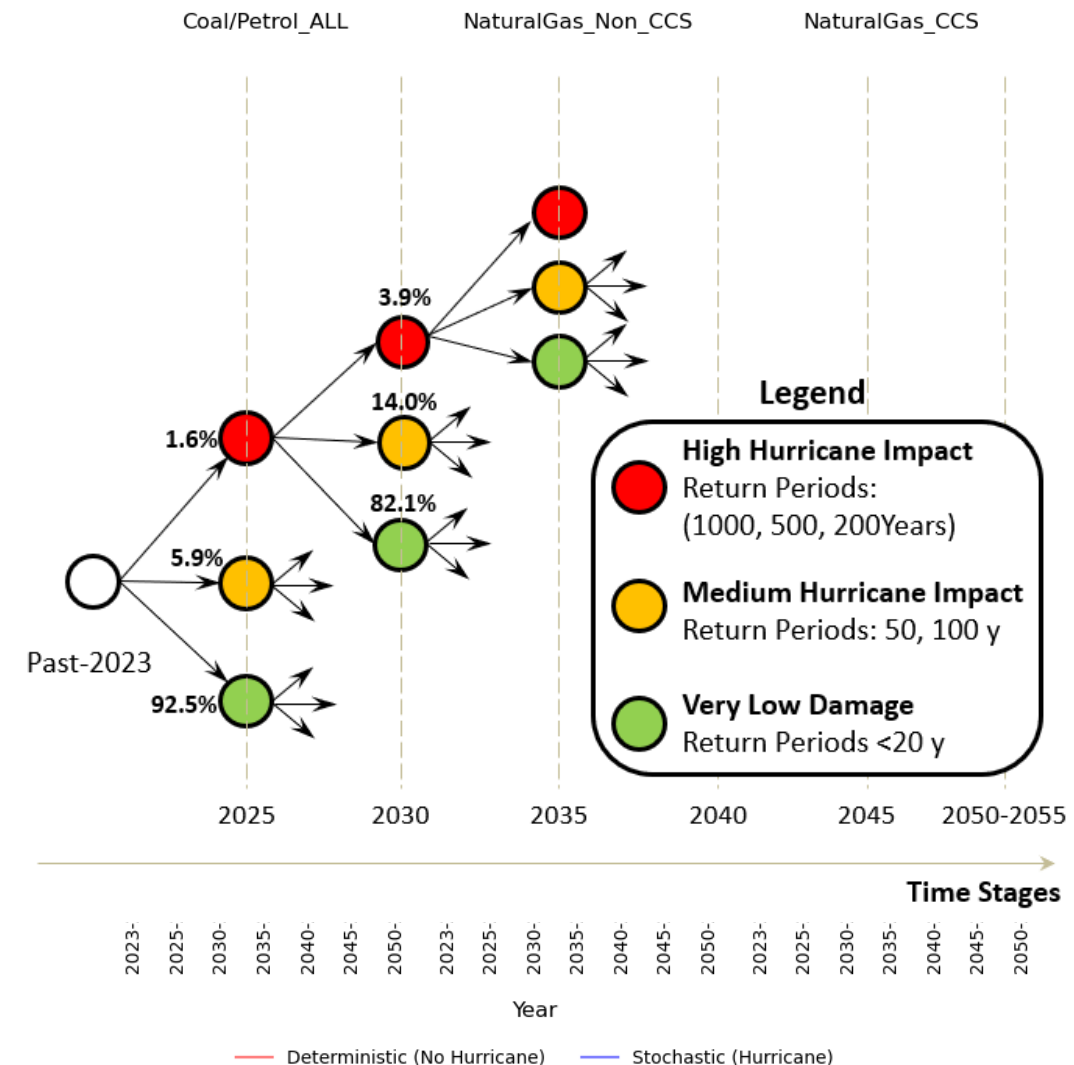


# Stochastic Case

## Some Observations:

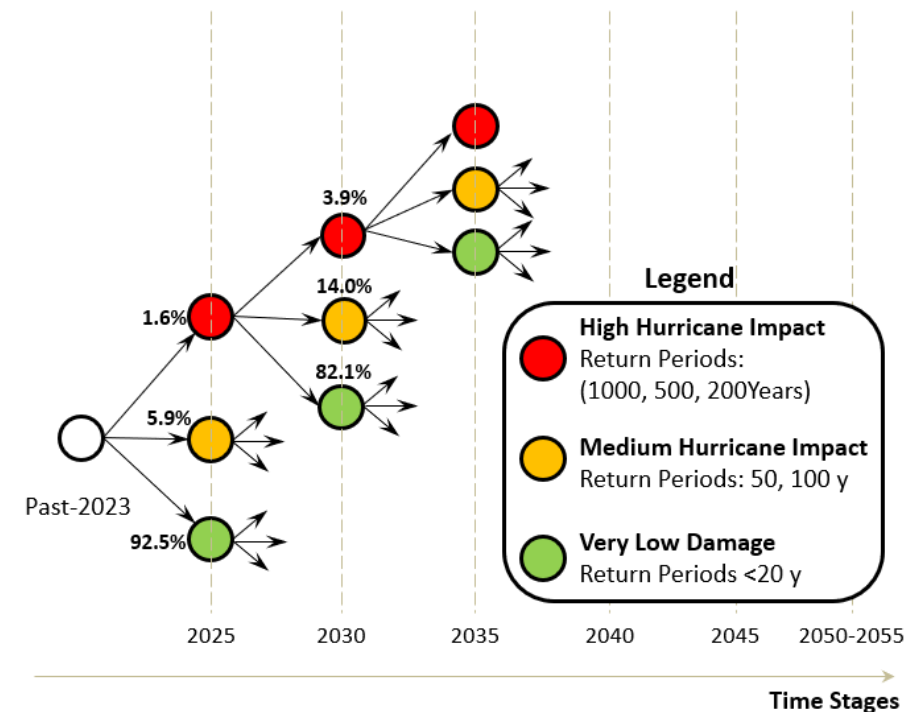
- Simulating **conservative case** with CO2 Targets, 3 scenarios per stage, 6-time stages (729 Scenarios), on a reduced number of representative hours per season.
- We did **not notice** any **substantial changes** between the **deterministic (No Hurricane)** and **stochastic models** regarding the activity of individual technology groups
- 0.4% increase in the OBJECTIVE costs** from the stochastic model in comparison with the deterministic (No Hurricane)

Fig 9. Total Activity for Different Technologies Under Conservative Cost Assumptions, With CO2 Targets (Stoch Wind Speed +15%, Deterministic Model With No Hurricane)



# Final Remarks

- This is still a work in progress
- We have some ideas to better represent damage to infrastructure that could lead to changes in our model results.
  - Modeling damage to infrastructure as being realized during the middle of the time steps could lead to different energy reserve requirements.
- Currently, we assume damage is only realized at the end of each time step.
- Changes are expected in the stochastic model, but after those are concluded, we hope to also investigate other regions like Florida.



# Thank You!



## Any Questions?

### Our Contact Information:

Victor Faria ([vaduraes@ncsu.edu](mailto:vaduraes@ncsu.edu))

Matteo Nicoli ([matteo.nicoli@polito.it](mailto:matteo.nicoli@polito.it))

Anderson de Queiroz ([ardequei@ncsu.edu](mailto:ardequei@ncsu.edu))

